THE CHALLENGE Wearable video cameras with 360 degree panoramic view are an accentual tool for a police officer surrounded by a view rich with details. However, 360 videos requires novel applications for streaming and playback. Currently, several conventional applications are used, with different UI approaches, but unfortunately, with significant disadvantages in the public safety domain.
The figures below show currently used approaches.

Figure a: Rectangular view shows all content as a single warped image

Figure b: In dual view , a user needs to shift his attention between frontal and back camera feeds

Figure c: A user manually adjusts view area using UI controls (top left)

THE SOLUTION The AutoVista PoC solution use moving objects on a scene to form a cue for selecting and displaying important sections from the 360 video scenery.
The PoC addresses challenges like: fast motion estimation, handling of multiple objects of interest, smooth shift of attention between focus regions etc.
The resulting efficient implementation of the algorithms allows deployment of the algorithms directly to the edge device and, thus, additional valuable use cases.

WHAT'S NEXT? The next steps might include detection of additional cues for scene importance based on object recognition techniques (for instance, people or vehicle recognition) and usage of a laser pointer for interactive camera control.
AutoVista PoC implementation is a result of collaboration between the Israel Innovation Center (IIC) and Israel R&D Center. The project explores advanced computer vision techniques for automatic selection of the most interesting regions of 360 degree video content.

ALEX APARSTIN

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